SE LAB 5

NAME: SANCHITA V G
SRN: PES2UG24CS439
GIT LINK: https://github.com/sanchitavg/PES2UG24CS439_food_delivery_SE_LAB5

BEFORE:

```
(venv) sanchitaganesan@Sanchitas-MacBook-Air PES2UG24CS439_food_delivery_SE_LAB5 % clear; pylint delivery_system.py
************* Module delivery_system
delivery_system.py:1:0: C0114: Missing module docstring (missing-module-docstring)
delivery_system.py:7:0: C0103: Constant name "next_id" doesn't conform to UPPER_CASE naming style (invalid-name)
delivery_system.py:11:0: C0116: Missing function or method docstring (missing-function-docstring)
delivery_system.py:11:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
delivery_system.py:12:4: W0603: Using the global statement (global-statement)
delivery_system.py:21:0: C0116: Missing function or method docstring (missing-function-docstring)
delivery_system.py:24:4: W0702: No exception type(s) specified (bare-except)
delivery_system.py:27:0: C0116: Missing function or method docstring (missing-function-docstring)
delivery_system.py:28:4: W0603: Using the global statement (global-statement)
delivery_system.py:32:11: W0718: Catching too general exception Exception (broad-exception-caught)
delivery_system.py:30:12: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
delivery_system.py:30:12: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
delivery_system.py:35:0: C0116: Missing function or method docstring (missing-function-docstring)
delivery_system.py:42:0: C0116: Missing function or method docstring (missing-function-docstring)
delivery_system.py:43:8: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
delivery_system.py:47:0: C0116: Missing function or method docstring (missing-function-docstring)
delivery_system.py:48:4: W0603: Using the global statement (global-statement)
delivery_system.py:49:25: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
delivery_system.py:51:0: C0116: Missing function or method docstring (missing-function-docstring)
delivery_system.py:51:0: C0103: Function name "printQueue" doesn't conform to snake_case naming style (invalid-name)
delivery_system.py:52:10: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
delivery_system.py:56:0: C0116: Missing function or method docstring (missing-function-docstring)

-------------------------------------------------------------------
Your code has been rated at 6.27/10 (previous run: 6.27/10, +0.00)
```

Pylint rates the original code 6.27/10 with 22 messages, including missing docstrings, global statements, a bare except and an unclosed file.

```
(venv) sanchitaganesan@Sanchitas-MacBook-Air PES2UG24CS439_food_delivery_SE_LAB5 % clear; bandit -r delivery_system.py
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.14.6
Run started:2026-10-14 08:31:03.588391+00:00

Test results:
>> Issue: [B403:blacklist] Consider possible security implications associated with pickle module.
   Severity: Low   Confidence: High
   CWE: CWE-502 (https://cwe.mitre.org/data/definitions/502.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_imports.html#b403-import-pickle
   Location: ./delivery_system.py:2:0
1       import json
2       import pickle
3       from datetime import datetime

--------------------------------------------------
>> Issue: [B110:try_except_pass] Try, Except, Pass detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b110_try_except_pass.html
   Location: ./delivery_system.py:24:4
23              orders[order_id]["status"] = status
24          except:
25              pass
26

--------------------------------------------------
>> Issue: [B301:blacklist] Pickle and modules that wrap it can be unsafe when used to deserialize untrusted data, possible secu
rity issue.
   Severity: Medium   Confidence: High
   CWE: CWE-502 (https://cwe.mitre.org/data/definitions/502.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b301-pickle
   Location: ./delivery_system.py:49:13
```

```
48          global orders
49          orders = pickle.load(open(file, "rb"))
50

--------------------------------------------------

Code scanned:
        Total lines of code: 59
        Total lines skipped (#nosec): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 2
                Medium: 1
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 3
Files skipped (0):
```

Bandit flags 3 issues in the original code: unsafe `pickle` use (B301 Medium, B403 Low) and a `try/except/pass` (B110 Low).

```
(venv) sanchitaganesan@Sanchitas-MacBook-Air PES2UG24CS439_food_delivery_SE_LAB5 % clear; flake8 delivery_system.py
delivery_system.py:11:1: E302 expected 2 blank lines, found 1
delivery_system.py:16:80: E501 line too long (96 > 79 characters)
delivery_system.py:21:1: E302 expected 2 blank lines, found 1
delivery_system.py:24:5: E722 do not use bare 'except'
delivery_system.py:27:1: E302 expected 2 blank lines, found 1
delivery_system.py:35:1: E302 expected 2 blank lines, found 1
delivery_system.py:42:1: E302 expected 2 blank lines, found 1
delivery_system.py:47:1: E302 expected 2 blank lines, found 1
delivery_system.py:51:1: E302 expected 2 blank lines, found 1
delivery_system.py:54:80: E501 line too long (80 > 79 characters)
delivery_system.py:56:1: E302 expected 2 blank lines, found 1
delivery_system.py:70:1: E305 expected 2 blank lines after class or function definition, found 1
```

Flake8 reports 12 PEP 8 violations in the original code: blank-line errors (E302, E305), long lines (E501) and a bare `except` (E722).

AFTER:

```
(venv) sanchitaganesan@Sanchitas-MacBook-Air PES2UG24CS439_food_delivery_SE_LAB5 % bandit -r delivery_system.py
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.14.6
Run started:2026-10-14 08:35:28.476610+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 63
        Total lines skipped (#nosec): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
```

After replacing pickle with JSON and removing `try/except/pass`, Bandit reports 0 issues at every severity level.

```
************* Module delivery_system
delivery_system.py:74:4: C0116: Missing function or method docstring (missing-function-docstring)


-------------------------------------------------------------------
Your code has been rated at 9.86/10 (previous run: 9.86/10, +0.00)
```

After the fixes Pylint rates the code 9.86/10, with only one minor missing-docstring message left.

```
(venv) sanchitaganesan@Sanchitas-MacBook-Air PES2UG24CS439_food_delivery_SE_LAB5 % flake8 delivery_system.py
(venv) sanchitaganesan@Sanchitas-MacBook-Air PES2UG24CS439_food_delivery_SE_LAB5 %
```

After fixing the formatting, Flake8 prints nothing, meaning there are no PEP 8 violations.